

Algorithmic Trading

INTRODUCTION

Algorithmic trading represents the computerized executions of financial instruments. Algorithms trade stocks, bonds, currencies, and a plethora of financial derivatives. Algorithms are also fundamental to investment strategies and trading goals. The new era of trading provides investors with more efficient executions while lowering transaction costs—the result, improved portfolio performance. Algorithmic trading has been referred to as “automated,” “black box” and “robo” trading.

Trading via algorithms requires investors to first specify their investing and/or trading goals in terms of mathematical instructions. Dependent upon investors’ needs, customized instructions range from simple to highly sophisticated. After instructions are specified, computers implement those trades following the prescribed instructions.

Managers use algorithms in a variety of ways. Money management funds—mutual and index funds, pension plans, quantitative funds and even hedge funds—use algorithms to implement investment decisions. In these cases, money managers use different stock selection and portfolio construction techniques to determine their preferred holdings, and then employ algorithms to implement those decisions. Algorithms determine the best way to slice orders and trade over time. They determine appropriate price, time, and quantity of shares (size) to enter the market. Often, these algorithms make decisions independent of any human interaction.

Similar to a more antiquated, manual market-making approach, broker dealers and market makers now use automated algorithms to provide liquidity to the marketplace. As such, these parties are able to make markets in a broader spectrum of securities electronically rather than manually, cutting costs of hiring additional traders.

Aside from improving liquidity to the marketplace, broker dealers are using algorithms to transact for investor clients. Once investment decisions are made, buy-side trading desks pass orders to their brokers for execution